# OCC: SOFTWARE ENGINEER YEAR 2

## SAQA ID: 119458

## NQF Level: 6

## Software Design & Testing with C#

## SDT621

## Formative Assessment 1

## 2026

CTU training solutions | Private Tertiary Education

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# DECLARATION OF AUTHENTICITY

I Nhlanhla Sibiya (FULL NAME)

hereby declare that the contents of this assessment SDT621 FA1

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

I have read and understood the Examination Rules and Regulations.

Signature: _____

Date: 27 April 2026



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# QUALIFICATION AND ASSESSMENT DETAILS

| | |
|---|---|
| **Programme Title** | OCC: SOFTWARE ENGINEER |
| **Module Title** | Software design and testing with C# |
| **Module Code** | SDT621 |
| **NQF Level** | 6 |
| **Assessment Type** | Formative Assessment 1 |
| **Hand Out Date** | 13 / 0 4 / 2026 |
| **Hand in Date** | 28 / 04 / 2026 |
| **Total Marks** | 100 |
| **Assessment Developer** | Lusukama Selemani |
| **Internal Moderator** | Faith Muwishi |
| **Internal Moderator** | Michelle Du Toit |
| **External Moderator** | |

| | |
|---|---|
| **Student Name and Surname** | Nhlanhla Sibiya |
| **Student Number** | 20250441 |


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# Instructions:

Please read the following instructions carefully before attempting the assessment:

- Read each question carefully and consider the mark allocation before answering.
- Ensure that you answer all questions and provide clear, well-structured C# code and explanations where required.
- Apply your understanding of C# programming concepts to solve the problems based on the given scenarios.

# Submission Requirements

- For your final submission, you must include:
- The Declaration of Authenticity
- Your completed Answer Sheet (PDF document)
- Your Visual Studio project files (compressed as a ZIP file)

# Evidence Collection Requirements

- To support your submission, you must:
- Take a screenshot of the output for each question
- Ensure each screenshot shows the full screen, including the system date and time
- Do NOT submit screenshots of your source code
- Submit your complete project folder as a ZIP file

# GitHub Submission Requirement

You are required to:

- Push your project code to GitHub
- Ensure the repository contains the correct and complete solution
- Share the GitHub repository link in your submission for review and verification
- Example: https://github.com/your-username/project-name

**Please ensure that your repository:**

- builds successfully
- contains all project files
- matches the work submitted in your ZIP file


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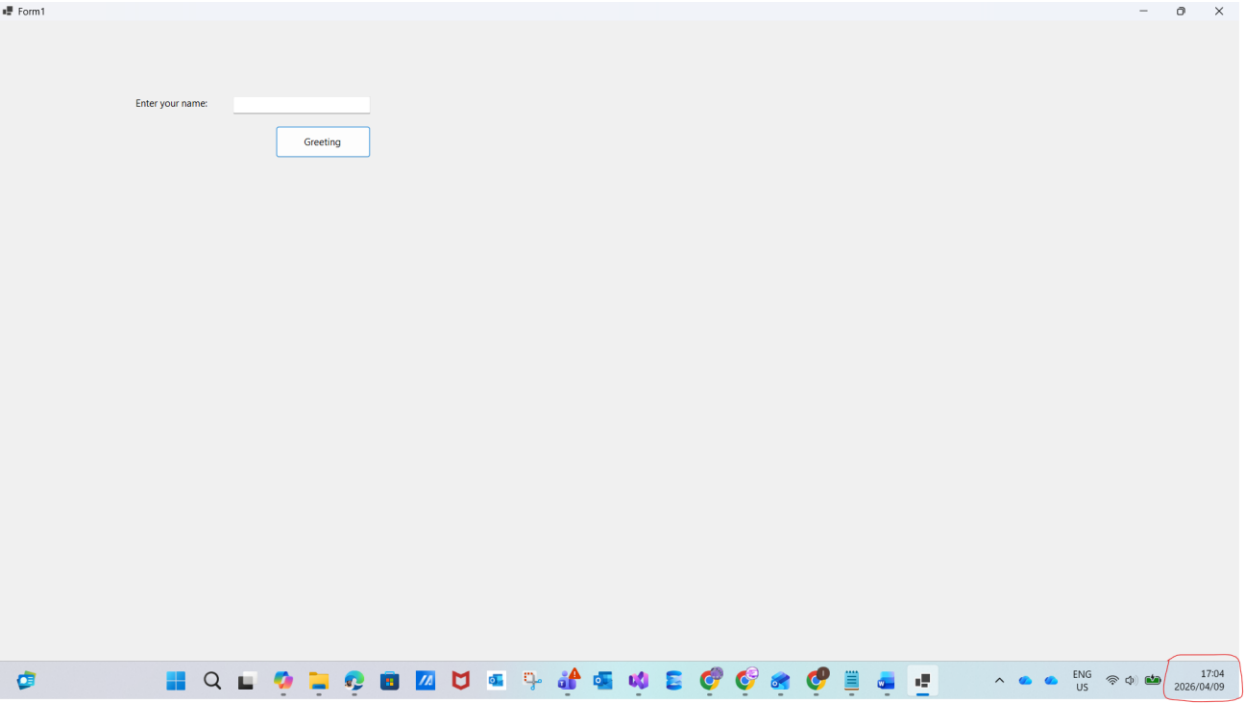
# Assessment Outcomes:

- Create and run C# programs in Visual Studio.
- Declare and manipulate variables.
- Use operators and expressions.
- Understand scope (local, class-level, etc.).
- Write methods with parameters and return values.
- Create classes, objects, fields, properties, and constructors.

## Example of Evidence Collection for Your Submission (PDF):

SECTION A: Short Answer Questions — 40 Marks

Question1 :




Ensure each screenshot shows the **full screen**, including the **system date and time**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## SECTION A: Short Answer Questions. 40 Marks

## Question 1 10 Marks

Write a C# console application program that produces output similar to the screenshot provided.

```
C:\Users\LusukamaSelemani\:

Enter student name: Amina
Enter mark for Subject 1: 98
Enter mark for Subject 2: 96
Enter mark for Subject 3: 87

===== STUDENT RESULTS =====
Student Name: Amina
Total Marks: 281
Average Marks: 93,7
Result: PASS
Result Issued At: 09 Apr 2026 15:32:56

Press any key to exit...
```

Your program must:

- Prompt the user to enter a student name
- Prompt the user to enter three subject marks
- Validate that marks entered are numeric values
- Calculate the total marks
- Calculate the average marks
- Display whether the student PASS or FAIL
- Rule: Average ≥ 50 = PASS
- Display results in the format shown above

## Question 2 10 marks

Develop a Simple ATM Console Application using C# that simulates a basic banking withdrawal transaction. The system must interact with the user, accept inputs, perform calculations, and display transaction results including the updated balance and transaction time.

```
C:\Users\LusukamaSelemani\:

===== CTU SIMPLE ATM SYSTEM =====

HI , WHAT IS YOUR NAME?
Salama

WELCOME SALAMA!
Enter account balance: 100000000
Enter withdrawal amount: 567432

Withdrawal successful!
Updated Balance: 99432568,00
Transaction Time: 09 Apr 2026 15:43:44
```


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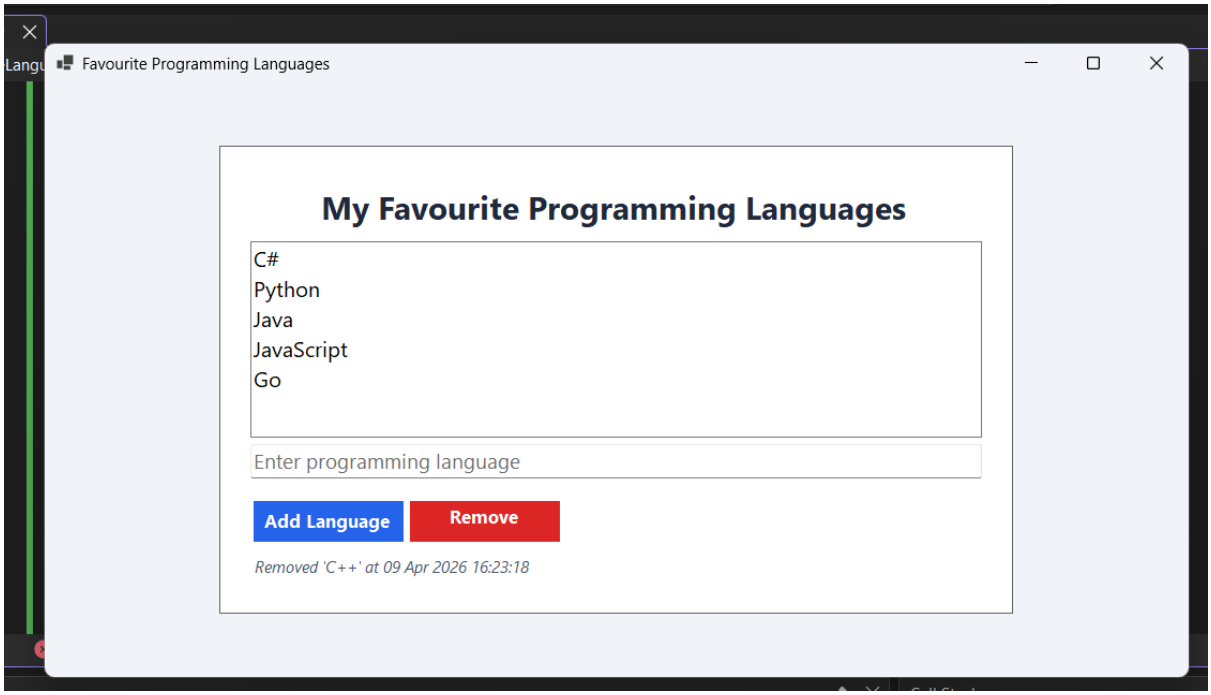
## Question 3 20 Marks

Create a C# Windows Forms App application that performs the following functions:

- Allow the user to add a programming language to the list
- Prevent empty input from being added
- Prevent duplicate languages from being added
- Allow the user to remove a selected language
- Display the date and time when a language

Expected output:





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## SECTION B: Long Answer Questions. 50 Marks

## Question 1: Emfuleni Municipality Console Application (20 Marks)

### Scenario

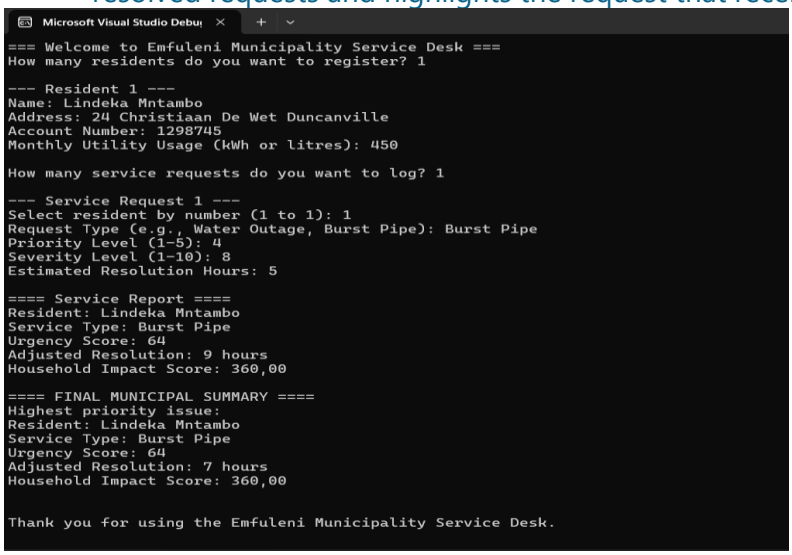
You are required to design and implement a console-based municipal service management application for Emfuleni Municipality. The purpose of this system is to enable the municipality to capture resident information, record utility-related service requests, and manage the processing of these requests interactively according to their priority and urgency levels.

### Class Design

- **Resident.cs:** This class is responsible for storing resident information. The attributes include name, address, account number, and monthly utility usage.
- **ServiceRequest.cs:** This class captures and stores details about service requests. The attributes include request type, priority level, severity level, and estimated resolution time.
- **UtilitiesManager.cs:** This class processes service requests. Its responsibilities include calculating urgency scores for requests and generating comprehensive service reports.

## Main Program Workflow (Program.cs)

1. The system prompts the user to enter the number of residents and gathers each resident's details.
2. Next, the user is asked to specify the number of service requests, after which the system collects information for each request. This includes associating the request with a specific resident, capturing the request type, priority level (ranging from 1 to 5), severity level (ranging from 1 to 10), and estimated resolution hours.
3. The application displays a queue of pending service requests, each accompanied by its urgency score.
4. The user can interactively select requests from the queue to process.
5. For each processed request, the application generates a report that includes resident details, service request details, and the calculated urgency score.
6. After all requests have been processed, a summary is shown. This summary lists all resolved requests and highlights the request that received the highest urgency score.



```
Microsoft Visual Studio Debu
=== Welcome to Emfuleni Municipality Service Desk ===
How many residents do you want to register? 1

--- Resident 1 ---
Name: Lindeka Mntambo
Address: 24 Christiaan De Wet Duncanville
Account Number: 1298745
Monthly Utility Usage (kWh or litres): 450

How many service requests do you want to log? 1

--- Service Request 1 ---
Select resident by number (1 to 1): 1
Request Type (e.g., Water Outage, Burst Pipe): Burst Pipe
Priority Level (1-5): 4
Severity Level (1-10): 8
Estimated Resolution Hours: 5

==== Service Report ====
Resident: Lindeka Mntambo
Service Type: Burst Pipe
Urgency Score: 64
Adjusted Resolution: 9 hours
Household Impact Score: 360,00

==== FINAL MUNICIPAL SUMMARY ====
Highest priority issue:
Resident: Lindeka Mntambo
Service Type: Burst Pipe
Urgency Score: 64
Adjusted Resolution: 7 hours
Household Impact Score: 360,00


Thank you for using the Emfuleni Municipality Service Desk.
```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## Question 2: Digital Identity Processor – Windows Forms Application 30 Marks

### Application Overview

The Windows Forms Application, named **HomeAffairsDigitalIdentityProcessor**, is designed to facilitate the creation and validation of digital citizen profiles. The application enables users to input a person's details, validate their identification number, and generate a summary profile in a graphical user interface.

### Graphical User Interface Components

- **Labels:** Used for guiding the user through the entry of name, ID number, citizenship selection, and for displaying results.
- **TextBoxes:** Two text boxes are provided; one for the user's name and one for the ID number.
- **ComboBox:** This control allows the user to select citizenship status from at least three options: Citizen, Permanent Resident, and Visitor.
- **Buttons:** There are two buttons—*Validate ID* and *Generate Profile*—to perform the respective actions.
- **Results Display:** The output summary is shown in a multi-line TextBox or Label, presenting the processed information.
- CitizenProfile Class Design
- The **CitizenProfile** class holds critical information for each profile and encapsulates the logic required for validation and age calculation.

### Properties:

- FullName
- IDNumber
- Age
- CitizenshipStatus
- **Constructor:** Accepts the name, ID number, and citizenship status, then calculates the age using a dedicated method.
- **Age Calculation:** The method extracts the birth date from the first six digits of the ID number (YYMMDD), determines the century (1900s or 2000s), and computes the current age accordingly.

### ValidateID() Method:

- Ensures the ID number contains exactly 13 digits.
- Verifies that the ID number is fully numeric.
- Uses the calculated age as part of the validation criteria.
- Returns a validation message indicating the result of these checks.


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### Application Functionality

- When the **Validate ID** button is pressed, the application reads the user inputs, creates a *CitizenProfile* object, and displays the validation result in the results area.
- Pressing the **Generate Profile** button produces a formatted profile summary containing the full name, ID number, calculated age, citizenship status, validation result, and a processing timestamp message.

Expected output

## External Evidence 10 Marks

GitHub link and zip file submitted with PDF answer.

**Missing link will result in losing 10 marks.**

TOTAL MARKS _____ / 100

**End of Assessment**


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## SECTION A RUBRIC (SHORT PRACTICAL QUESTIONS) – 40 MARKS

| Criteria | Excellent (Full Competency) | Competent (Partial Competency) | Needs Improvement | Marks |
|---|---|---|---|---|
| User Input Handling | Accepts and validates all inputs correctly | Accepts inputs but limited validation | Missing validation or incorrect input logic | 5 |
| Console Interaction Flow | Logical prompts and structured execution | Minor flow issues | Poor interaction structure | 5 |
| Calculation Logic | Accurate totals / averages / balances | Minor calculation errors | Incorrect calculations | 5 |
| Decision Structures | Correct PASS/FAIL and withdrawal logic | Minor logic errors | Missing or incorrect decisions | 5 |
| Output Formatting | Output matches required structure clearly | Minor formatting issues | Poor formatting | 5 |
| Windows Forms GUI Controls | Controls correctly implemented | Some missing controls | Incorrect GUI implementation | 5 |
| Event Handling | All events functional | Partial functionality | Events missing or incorrect | 5 |
| Validation & Defensive Programming | Prevents duplicates / empty input / invalid data | Partial validation | No validation | 5 |

Section A Total = _____ / 40 Marks

## SECTION B RUBRIC (LONG PRACTICAL QUESTIONS) – 50 MARKS

### QUESTION 1 RUBRIC

Emfuleni Municipality Console Application (20 Marks)

| Criteria | Excellent | Competent | Needs Improvement | Marks |
|---|---|---|---|---|


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| | | | | |
|---|---|---|---|---|
| Resident Class Implementation | All attributes correct with constructor & usage | Minor omissions | Class poorly structured | 3 |
| ServiceRequest Class Implementation | Fully captures request properties correctly | Minor attribute issues | Incorrect class design | 3 |
| UtilitiesManager Class Logic | Correct urgency calculation + reporting workflow | Partial processing logic | Missing manager logic | 4 |
| Resident Data Capture Workflow | Captures residents dynamically | Minor interaction issues | Static / incomplete capture | 2 |
| Service Request Capture Workflow | Captures requests correctly with associations | Minor linkage issues | Incorrect request handling | 2 |
| Priority + Severity Validation | Correct numeric validation ranges enforced | Partial validation | No validation | 2 |
| Queue Display with Urgency Scores | Displays structured queue correctly | Minor display issues | Queue missing | 1 |
| Interactive Processing Simulation | Allows request processing selection | Limited interaction | Missing interaction | 1 |
| Generated Processing Report | Report includes resident + request + urgency score | Partial report output | Report incomplete | 1 |
| Final Summary Report | Lists resolved requests + highest urgency | Partial summary | Missing summary | 1 |

Question 1 Total = _____ / 20 Marks

## QUESTION 2 RUBRIC

Digital Identity Processor – Windows Forms Application (30 Marks)

| Criteria | Excellent | Competent | Needs Improvement | Marks |
|---|---|---|---|---|
| GUI Layout Structure | All controls implemented correctly | Minor missing components | Incorrect layout | 4 |
| Label Usage & Guidance Structure | Labels clearly guide user workflow | Minor clarity issues | Poor UI guidance | 2 |
| TextBox Inputs (Name + ID) | Inputs captured correctly | Minor validation issues | Incorrect capture | 2 |


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| | | | | |
|---|---|---|---|---|
| ComboBox Citizenship Selection | Correct implementation with required options | Minor option issue | Incorrect setup | 2 |
| Validate ID Button Logic | Executes validation workflow correctly | Partial validation | Incorrect logic | 4 |
| CitizenProfile Class Design | Proper encapsulation with properties + constructor | Minor class issues | Incorrect structure | 4 |
| Age Calculation Logic (YYMMDD extraction) | Correct century detection & age calculation | Minor logic issues | Incorrect age logic | 4 |
| ValidateID() Method Implementation | Checks numeric + 13 digits + logical validation | Partial checks | Missing validation | 3 |
| Generate Profile Button Logic | Generates structured formatted profile summary | Partial formatting | Missing summary output | 3 |
| Timestamp Display | Timestamp correctly displayed | Minor formatting issue | Missing timestamp | 1 |
| Results Display Output | Clear structured multiline output | Minor formatting issues | Poor output formatting | 1 |

Question 2 Total = _____ / 30 Marks

## EXTERNAL EVIDENCE RUBRIC – 10 MARKS

This supports authenticity verification and workplace readiness expectations

| Criteria | Excellent | Competent | Needs Improvement | Marks |
|---|---|---|---|---|
| GitHub Repository Submitted | Repository accessible with full solution | Repository incomplete | Missing repository | 5 |
| Zip File Submitted | Full working solution included | Partial solution included | Missing zip file | 5 |

External Evidence Total = _____ / 10 Marks


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## MARK ALLOCATION

| Section | Description | Marks |
|---|---|---|
| Section A | Short Programming Tasks | _____ / 40 |
| Section B Question 1 | Municipality Console Application | _____ / 20 |
| Section B Question 2 | Windows Forms Identity Processor | _____ / 30 |
| External Evidence | GitHub + Zip Submission | _____ / 10 |
| TOTAL | | _____ / 100 Marks |

# LEARNING OUTCOMES CROSS-MAPPING

| Learning Outcome | Description | Question Alignment | Evidence Produced |
|---|---|---|---|
| LO1 | Apply variables and data types | Section A, B | Input capture |
| LO2 | Implement decision structures | Section A, B | PASS/FAIL, validation |
| LO3 | Perform numeric validation | Section A, B | Range checking |
| LO4 | Develop console applications | Section A, B Q1 | Workflow simulation |
| LO5 | Implement object-oriented classes | Section B Q1 & Q2 | Resident + Profile classes |
| LO6 | Apply encapsulation | Section B | CitizenProfile |
| LO7 | Implement event-driven programming | Section A Q3 + Section B Q2 | Button actions |
| LO8 | Perform data validation logic | Section A + B | Duplicate prevention |
| LO9 | Implement structured reports | Section B | Service summary |
| LO10 | Apply GUI development principles | Section A Q3 + Section B Q2 | WinForms design |
| LO11 | Apply algorithmic workflow modelling | Section B Q1 | Queue processing |
| LO12 | Implement defensive programming | Section A + B | Input validation |


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